

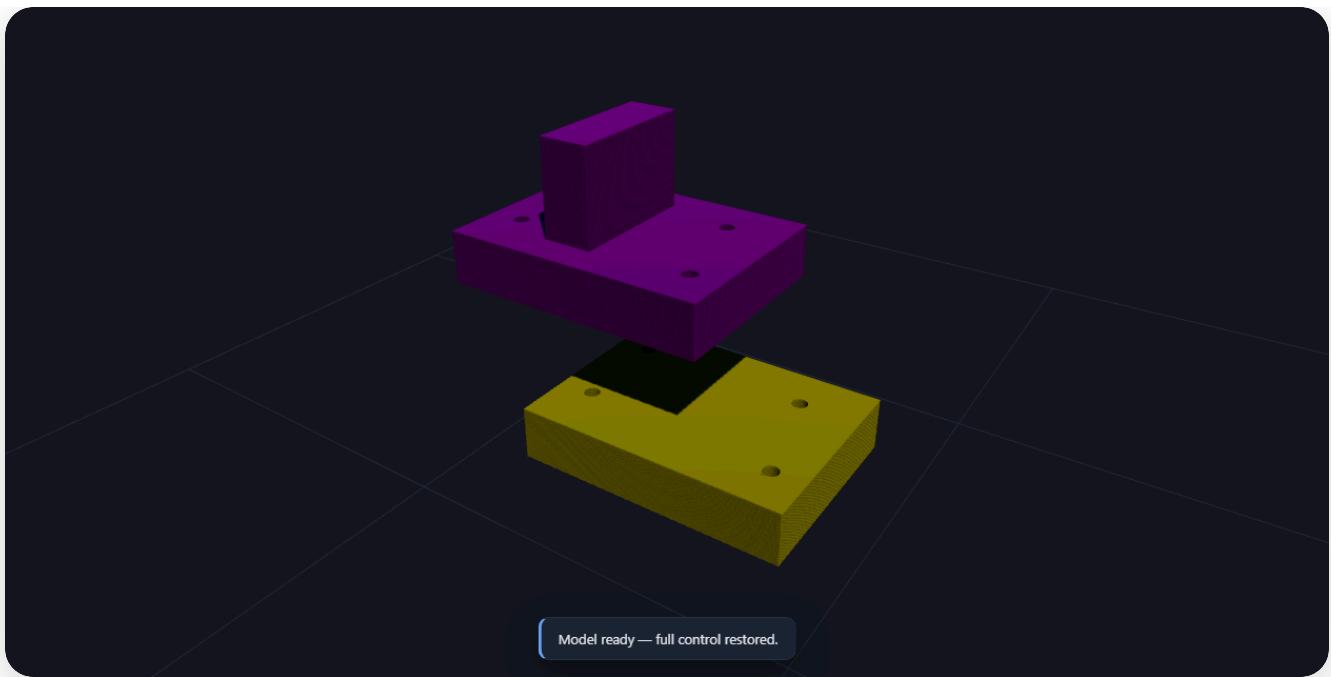
CAD Viewer — User Manual

Version: v0.55 · **URL:** <http://localhost:8088/>

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1. Welcome — why CAD Viewer?



Turn a CAD file into a live, shared 3D scene. CAD Viewer is a real-time collaborative browser app for viewing CAD geometry. Anyone on your network can open the model with nothing more than a web browser — no CAD software, no licences, no installs. Everyone shares the same view, and everyone can interact with it.

At a glance

Instead of emailing files back and forth, share a live model:

Instead of...	CAD Viewer lets you...
Export → email → wait → open → repeat	Share one join link ; the model appears instantly
Sending multi-MB CAD bundles	Convert to lightweight GLB/GLTF once and share a small file
Explaining with static screenshots	Point, move and zoom together on the live model
Guests installing CAD software	Guests open the link in any browser — nothing to install
Worrying about files leaving your PC	The model is streamed to memory only ; nothing is written to the guest's disk

Application features

► **Both host and guest can interact.**

This is not a one-way broadcast — anyone in a session can orbit, zoom and pan, show or hide parts, highlight a part, and even move parts along X/Y/Z, mirrored live in both directions.

► **Live sync.**

Camera, part visibility, selection, and lighting/animation are shared between all members in real time. Late joiners automatically receive the current model and view state.

► **No install for guests.**

A guest needs only a browser — open a link and the shared model appears. Nothing to download, nothing to set up, no CAD licence.

► **Nothing is stored on the guest.**

The model is streamed to the guest's memory (RAM) only and rendered in the browser; no file is written to their disk. Close the tab and it's gone — ideal for sensitive geometry or NDA work.

► **Efficient sharing.**

STEP, IGES and OBJ files are converted to

GLB/GLTF

— a lightweight, web-native format — on the server, so one compact file moves fast across the network instead of a bulky CAD bundle.

► **Cross-platform & faithful.**

Works in any modern browser on Windows, macOS, Linux and tablets; part names, colours and materials survive the conversion, so reviewers see the design the way it was authored.

What it helps with

Online calibration of engineering design. Because everyone shares the same live view and can interact with it, the design-review / calibration loop collapses from email-and-export cycles into one shared session — questions are answered immediately, on the actual geometry, by pointing and looking together.

Use cases

► **Design review / sign-off**

► **Remote inspection**

review the model together across offices or the shop floor instead of exchanging exports.

check a part or assembly before manufacturing, from anywhere on the network.

► **Live design changes**

everyone watches the part move as it happens.

► **Client demo**

show the design without installing CAD on the client's machine.

► **Vendors & subcontractors**

share the live model with a supplier to align on geometry, fit and tolerances before fabrication.

► **Cross-site calibration**

compare a real part against the shared model live, from a different location.

2. Starting the app

Portable zip (Windows, no installs):

1. Unzip `cad-viewer-portable.zip` anywhere.
2. Double-click `start.bat` . It launches the server and opens the browser.
3. The app is served at **`http://localhost:8088/`**

From source: `npm start` (or `node src/server.js`) in the project folder, then open the same URL.

Changing the port

The default port is **8088**. Any of these overrides it:

Method	Example
Launcher argument (zip)	<code>start.bat 5555</code>
<code>port.txt</code> file next to the launcher (<code>start.bat</code> / <code>start.sh</code>)	create <code>port.txt</code> containing <code>5555</code>

Method	Example
Environment variable	<pre>set PORT=5555 then npm start (Windows) / PORT=5555 npm start (macOS/Linux)</pre>

Precedence: command-line argument → `port.txt` → env `PORT` → default 8088.

The join link and `/ip` address always reflect the actual port, so guests don't need to know it.

STEP, IGES and OBJ files (.step/.stp, .igs/.iges, .obj) additionally need

Docker + the `chair-cq:local` OpenCascade image. GLB/GLTF files work without any of it.

Installing the STEP converter (Docker + OpenCascade)

To open **STEP / IGES / OBJ** files, the server needs a small converter built on the OpenCascade CAD kernel, which runs inside a **Docker** container. This is a **one-time** setup on the PC that runs the server — guests never need it.

Run `install-docker-opencascade.bat` (double-click it; it sits in the same folder as `start.bat`). It walks through everything:

Step	What the script does
1. Docker check	If Docker Desktop isn't installed, it installs it automatically via <code>winget</code> (~500 MB download) and starts it. You can also install it yourself from https://www.docker.com/products/docker-desktop/ and re-run.
2. Wait for engine	Waits until the Docker engine is running (first start can take a few minutes).
3. Build the image	Builds the <code>chair-cq:local</code> OpenCascade image from the bundled <code>Dockerfile</code> — first build downloads ~1 GB and can take 5–15 minutes .
4. Verify	Runs a quick check that the OpenCascade kernel is ready, so you know STEP conversion will work.

To install it manually (instead of the script):

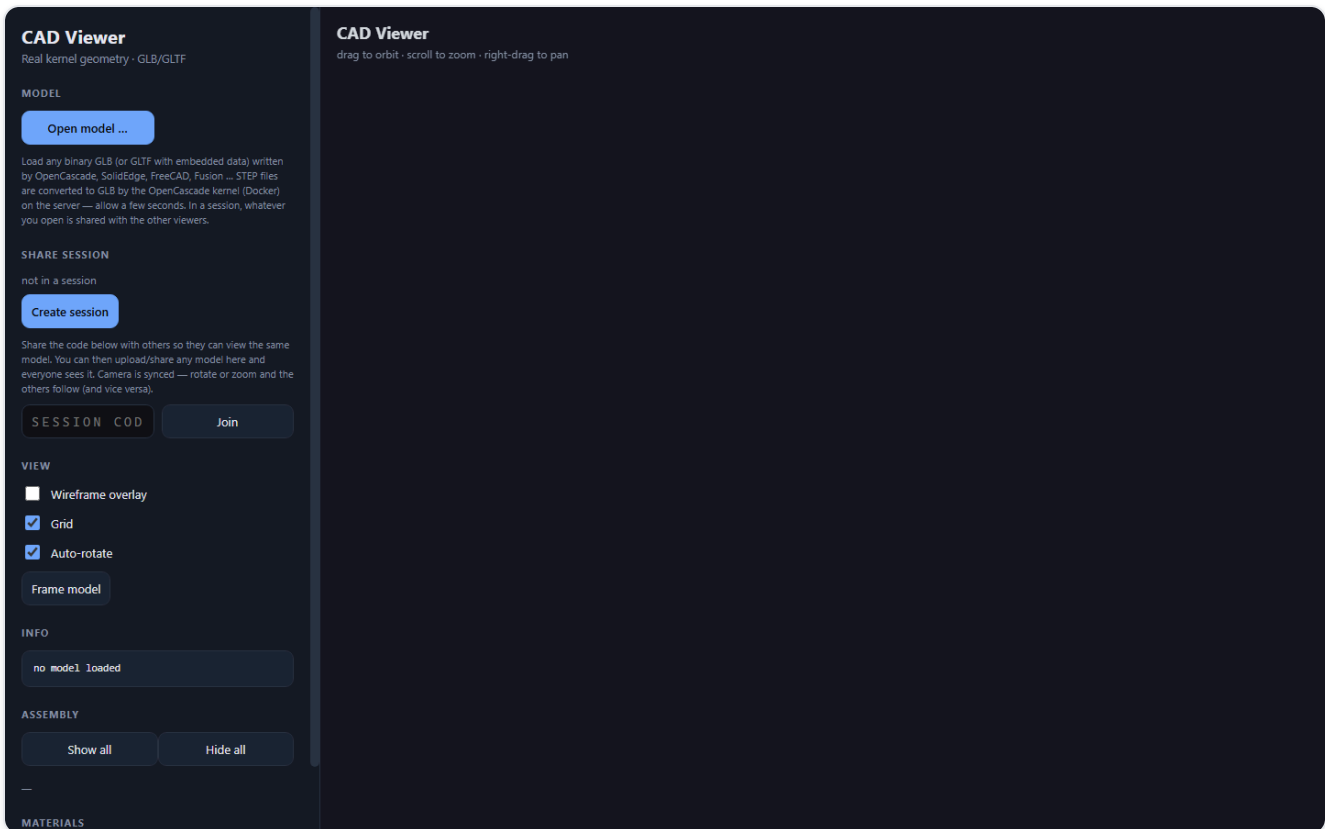
1. Install **Docker Desktop** and make sure the engine is running.
2. Open a terminal in the distribution folder and build the image:

```

```
docker build -t chair-cq:local .
```

```

That's it. Once the image exists, STEP / IGES / OBJ files convert normally. If you only ever open **GLB / GLTF** files, you can skip this entirely.

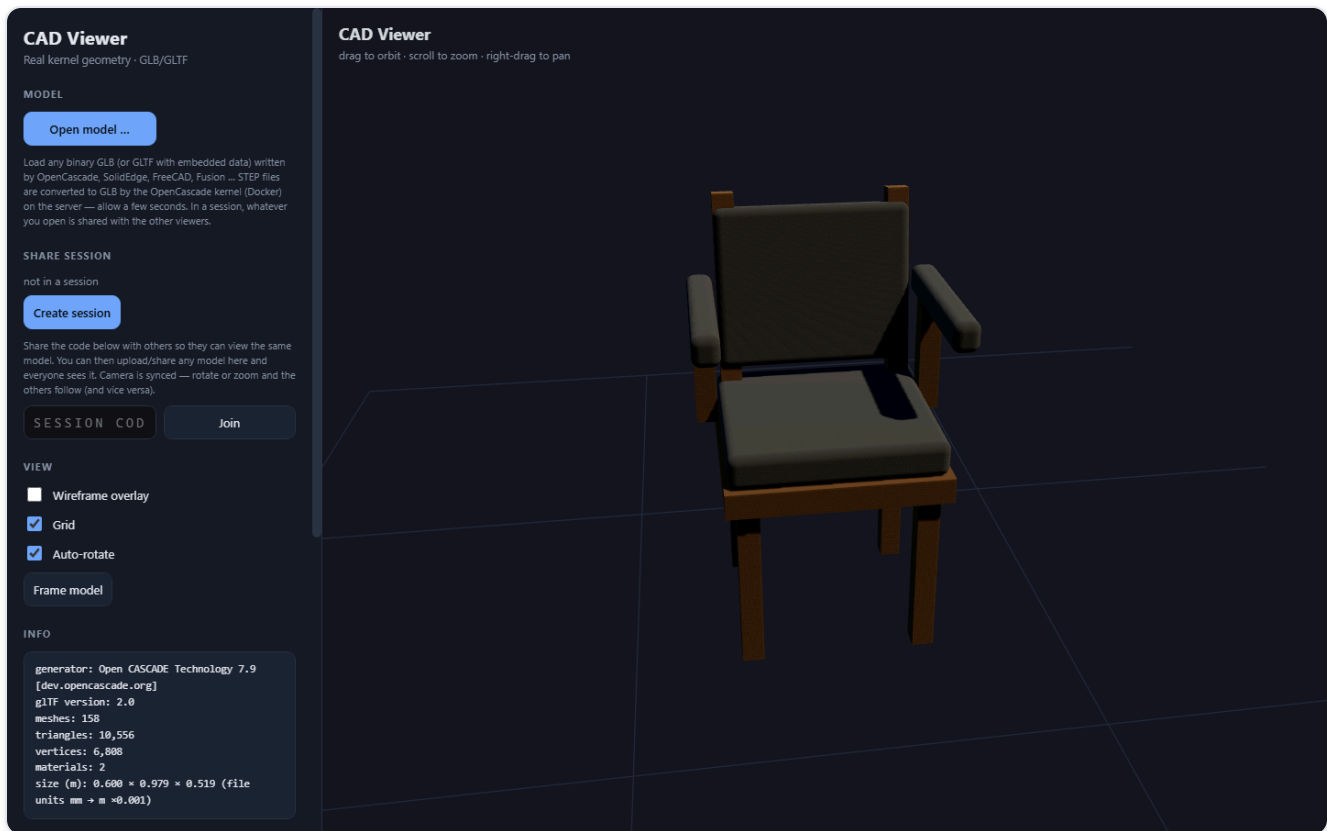


3. Opening a model

1. Click **Open model...** in the left sidebar.
2. Choose a `.glb`, `.gltf`, `.step`, `.stp`, `.igs`, `.iges` or `.obj` file from your computer.

GLB/GLTF loads instantly. STEP, IGES and OBJ files are converted to GLB by the

OpenCascade kernel on the server (a blocking overlay shows progress — a few seconds).

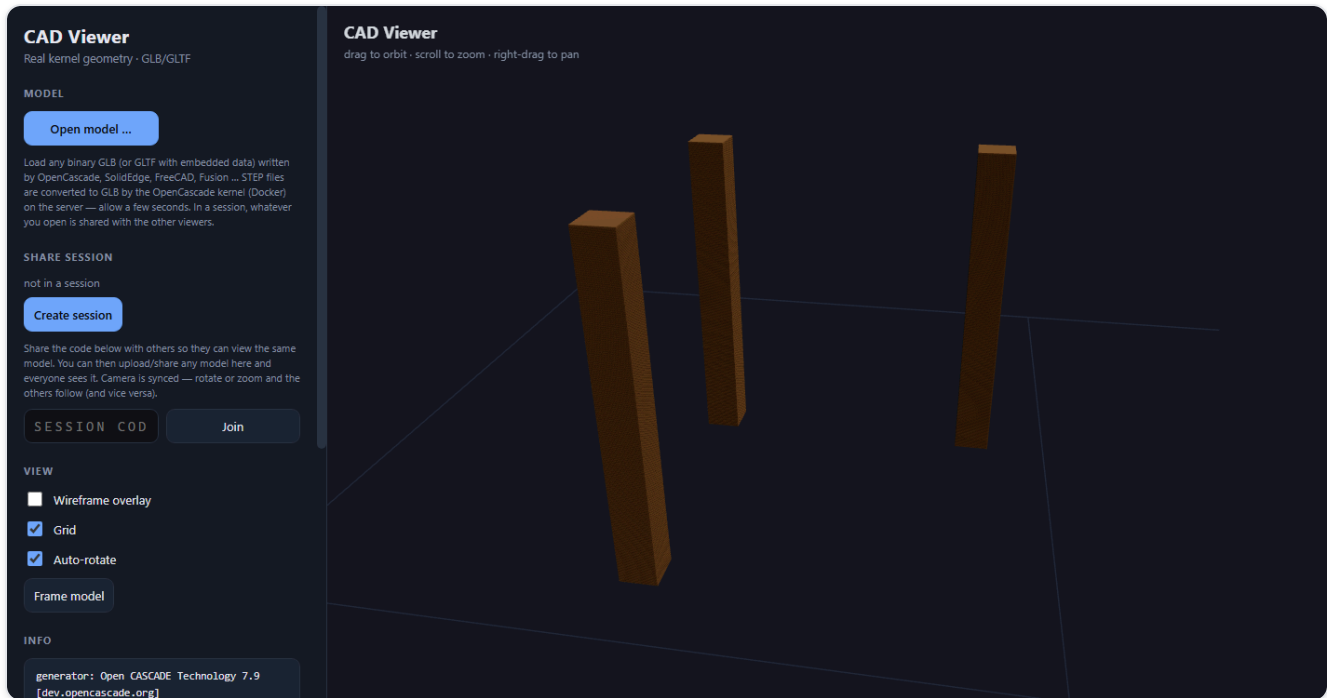


The sidebar shows you:

- **Info** — generator, mesh/triangle counts, dimensions and units
- **Assembly** — the part tree (see §4)
- **Materials** — swatches of every colour in the model

4. The Assembly panel — parts and visibility

Every part of the model appears in the **Assembly** tree. The tree opens expanded to **level 2** (the top assembly plus its direct children); deeper levels are collapsed until you expand them.



Expand / collapse

- Rows that contain sub-parts show a **+/- toggle** on the left.
- Click **+** to expand (show children), **-** to collapse (hide them).
- Expanding/collapsing is purely visual — it never changes part visibility.

Show / hide parts

- **Checkbox** per part: uncheck to hide it in the viewport, check to show.
- Toggling a parent cascades to its whole subtree — turning a sub-assembly on

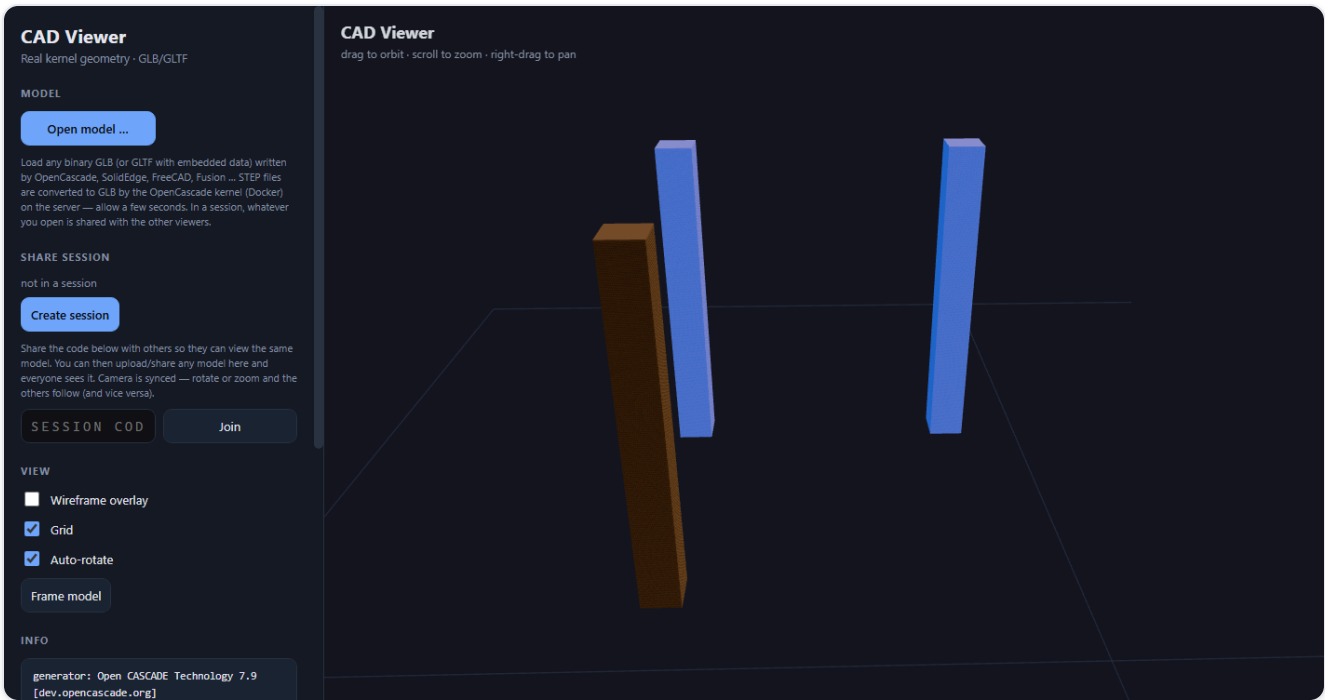
brings all its children back with it (three.js needs the whole chain visible to render anything).

- **Show all / Hide all** buttons apply to the whole model at once.

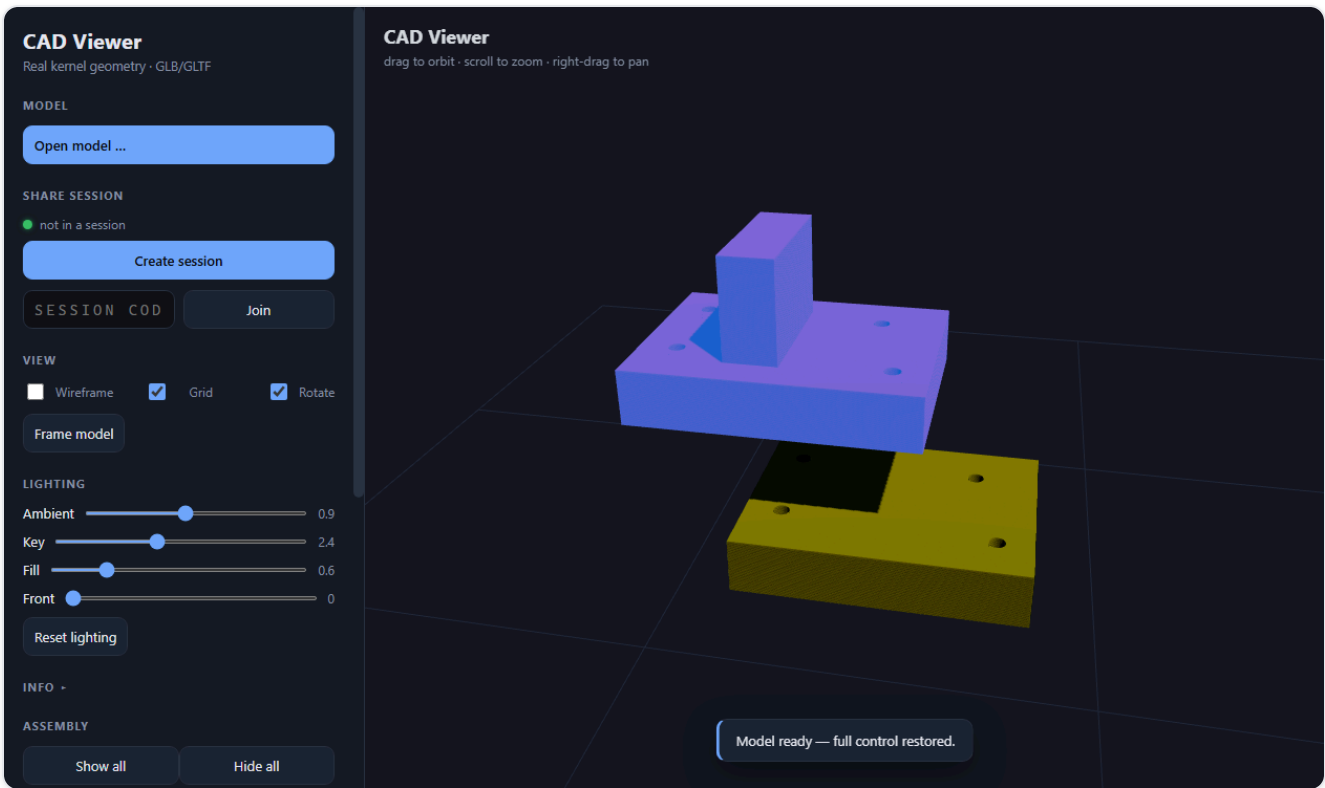
Highlight a part (click the name, or click it in the viewport)

Click a part's **name** to highlight it in blue in the viewport — or ****click the part directly in the 3D viewport**** to select it the same way (left-click on the part; left-click empty space to clear). Click another part to move the

highlight; click the same part again to clear it. The highlight uses per-mesh material copies, so other parts sharing the same source material are unaffected.

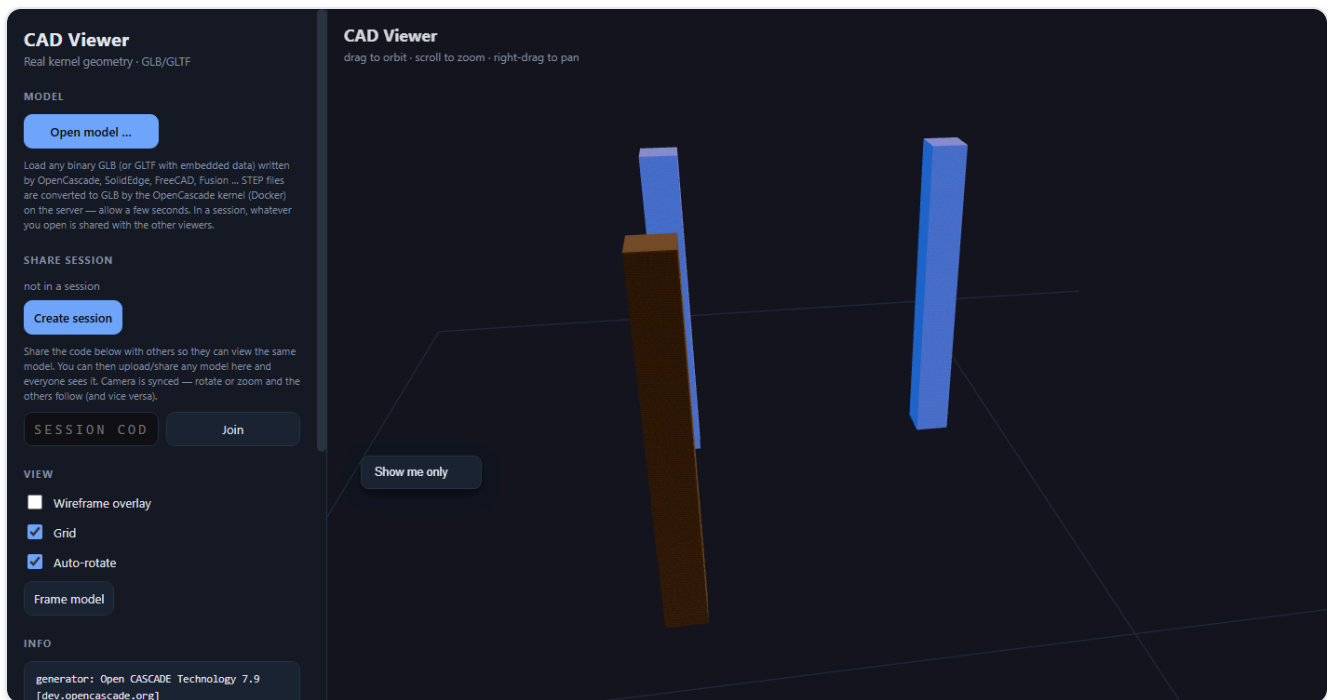


Clicking a part directly in the 3D viewport highlights it the same way:



"Show me only" (right-click)

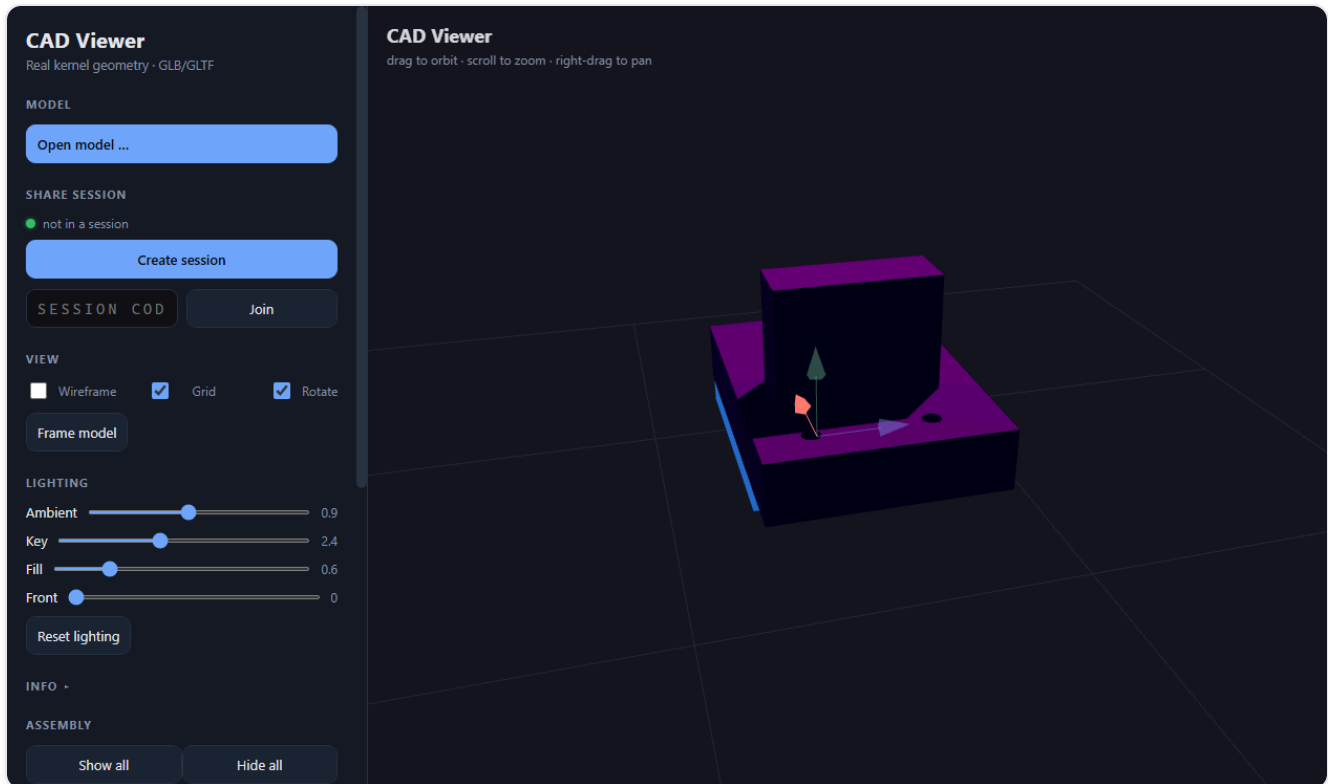
Right-click a part — either its name in the tree, or the part directly in the viewport — and choose **Show me only** to hide everything except that part and its children. Ancestors stay visible so the isolated part still renders.



Move a part (X / Y / Z + drag)

With a part highlighted, tick **Move part**, press **X**, **Y** or **Z** to choose an axis, then **left-drag** in the viewport to slide that part along the axis.

The highlighted **part moves by itself** — its siblings stay put. An X/Y/Z axis gizmo (red/green/blue) appears at the part so you can see the move directions; the chosen axis glows brighter. The gizmo stays about **1/8 of the screen** at any zoom level — zoom in and it shrinks to keep that size. **Undo** steps back through your last part movements (up to 200), and **Reset** returns all parts to their original positions and clears the move history. Moves are shared with the other viewers in the session.



In a session (see §7), all of this syncs to every member — show/hide, expand/collapse, the selection highlight, and part moves are mirrored live in both directions.

5. Navigating the viewport

Action	Effect
Drag (left button)	Orbit the camera
Scroll	Zoom
Right-drag	Pan
Frame model button	Reset the view to fit the model
Wireframe overlay	Toggle wireframe on all parts
Grid	Toggle the ground grid
Auto-rotate	Slow turntable spin (disabled in a session)

6. Lighting & animation

Lighting

A **Lighting** section in the sidebar lets you tune the light on the model:

Control	What it does	Range	Default
Ambient	Even fill from the sky and ground	0–2	0.9
Key	Main directional light (casts shadows)	0–6	2.4
Fill	Cool fill light from the opposite side	0–3	0.6
Front	Light straight from the viewer's side — handy when the front face is too dark	0–3	0 (off)

Drag a slider and the model updates live. **Reset lighting** restores the defaults.

Animation

If the GLB you load contains **keyframe animation** (e.g. from Blender or a game pipeline — CAD-converted STEP/IGES/OBJ files have none), an **Animation** section appears with:

Control	What it does
Play / Pause	Start or stop the clip
Loop	Repeat the clip (off = play once and stop)
Speed	Playback speed 0.25×–4×
Clip	Pick a clip when the model has several

Session sync

In a session, the **lighting levels (Ambient/Key/Fill/Front)** and the **animation state (clip, play/pause, loop, speed)** are shared with the other viewers, and late joiners receive the current settings. Camera, part visibility, tree expand/collapse and selection sync as well (see §8).

7. Collaborative sessions (host)

Sessions let other people on your LAN view the same model and follow your camera and part visibility.

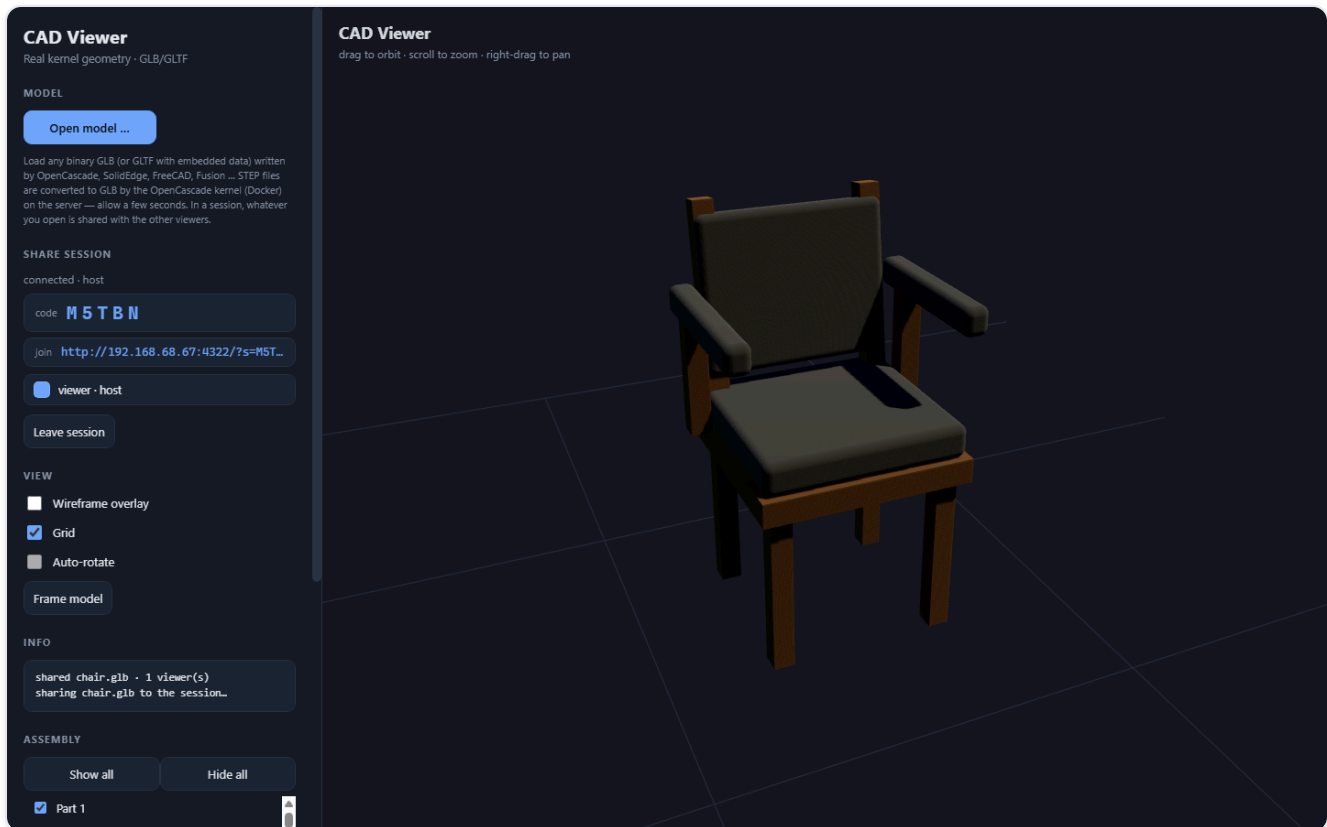
On launch you're asked to **enter your name**; it is shown to the other

viewers in the roster (e.g. Alice · host , Bob (you)). Your name is

remembered for next time.

1. Click **Create session** — a 5-character code appears.
2. **Open a model** (or re-share one you've already loaded).
3. Share the **join link** shown under the code, or the code itself, with the

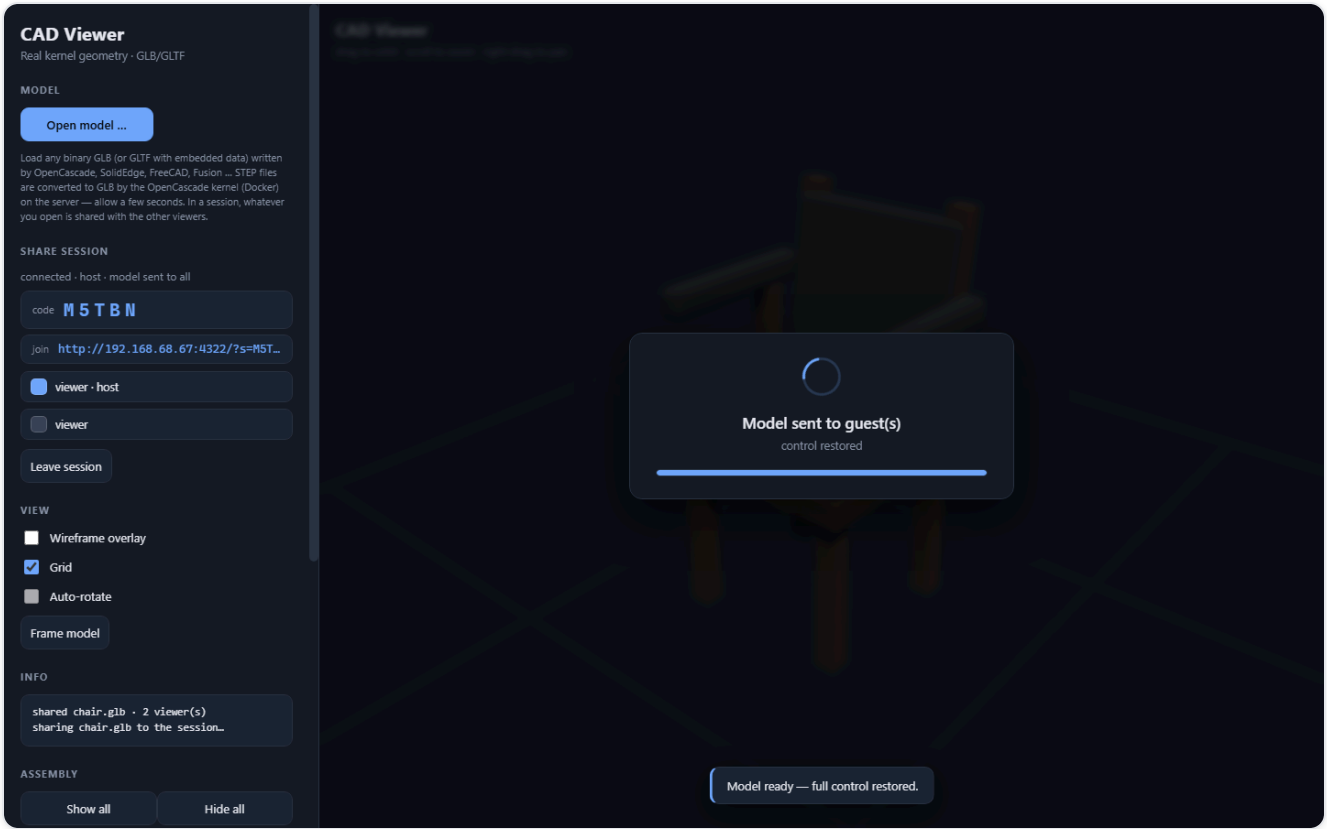
other users.



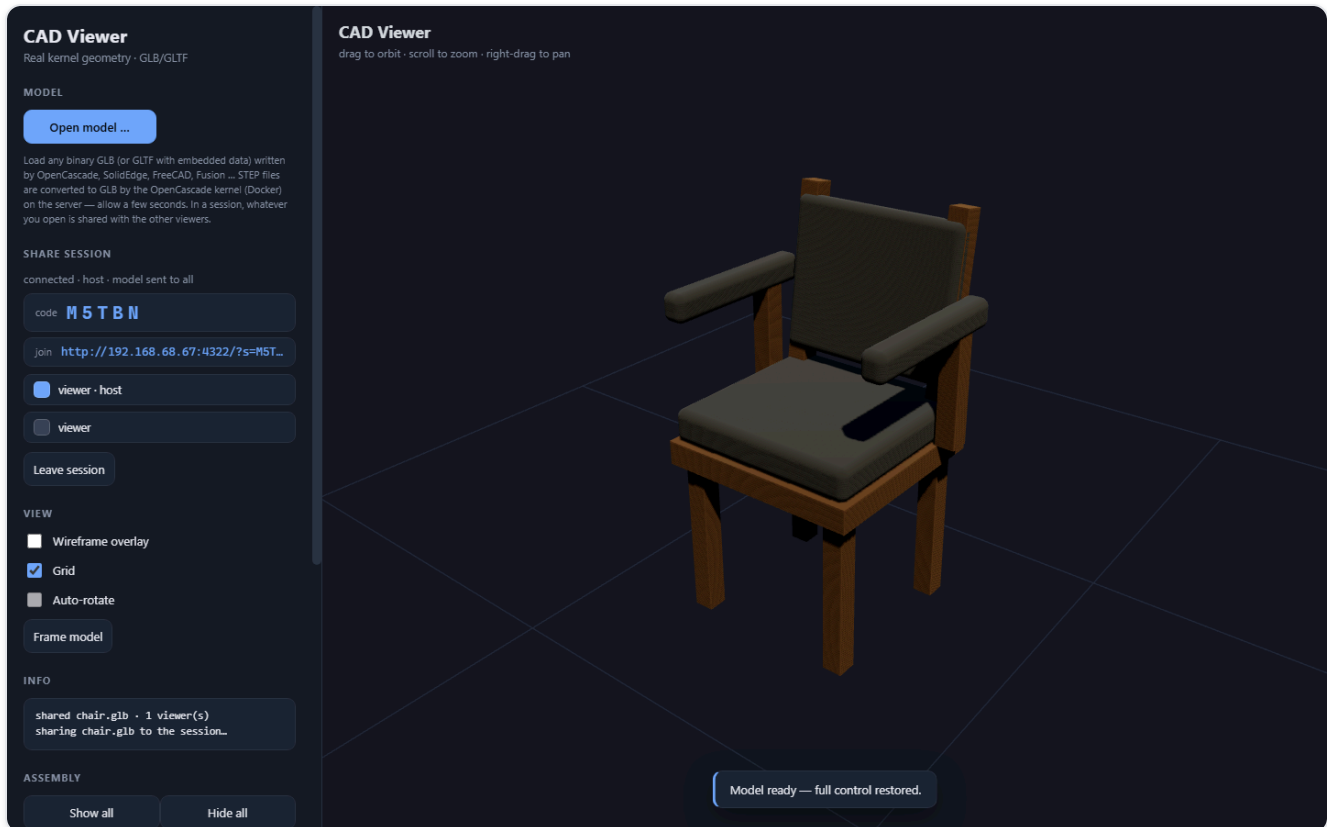
While the model is being handed to guests you'll see a blocking

“Sending model to guest(s)...” overlay; control is restored automatically once

every connected guest confirms receipt.



The roster under the code shows who is connected. The **host** can remove a viewer by clicking **kick** next to their name; the removed viewer's screen is cleared and they see *"removed by host"*. A small status dot next to "Share session" shows whether the server is reachable (green = up, red = down).

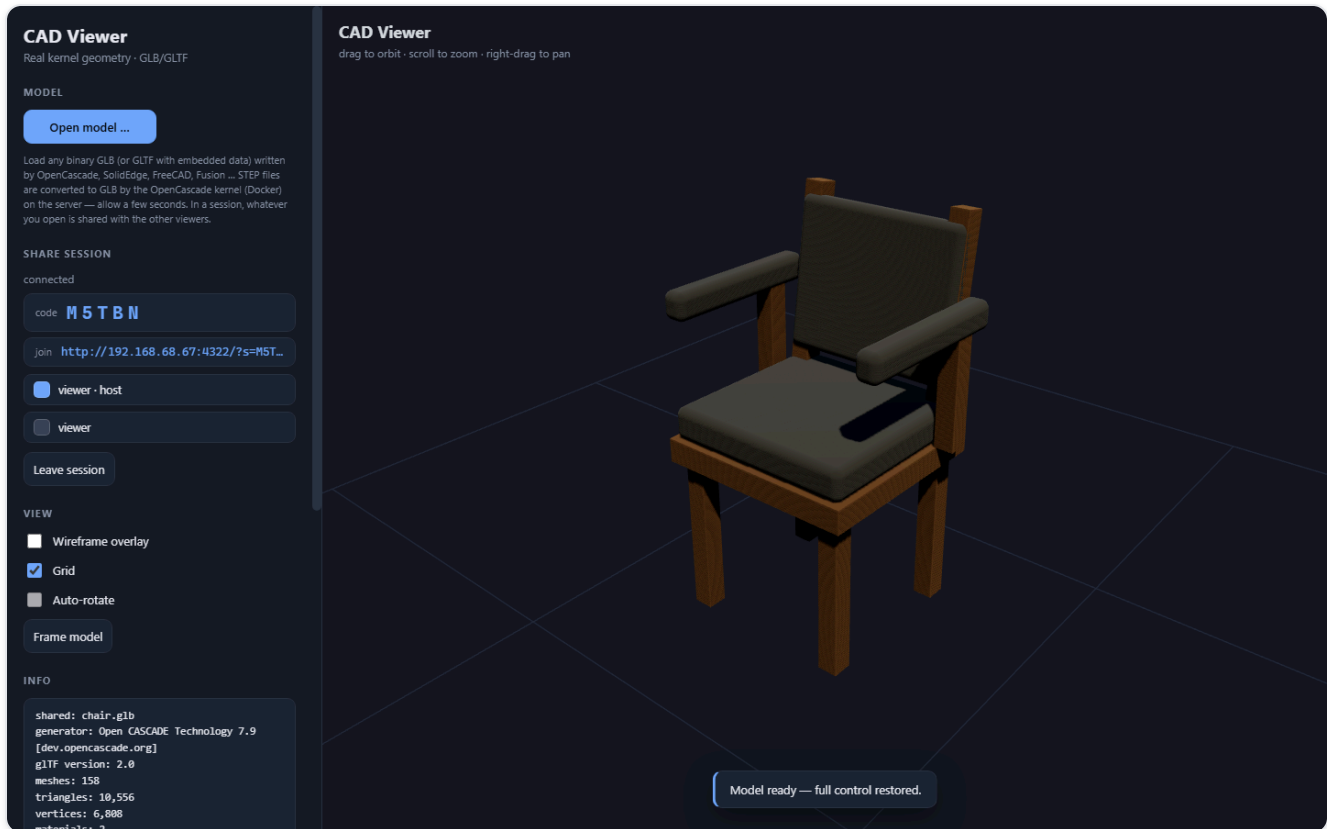


8. Joining a session (guest)

On another computer (same network):

1. Open the **join link** the host sent you, **or**
2. Open the app, type the code into **session code** and click **Join**.

You'll see a **“Receiving model...”** overlay with byte progress, then the model appears — same view, same parts, same visibility as the host.



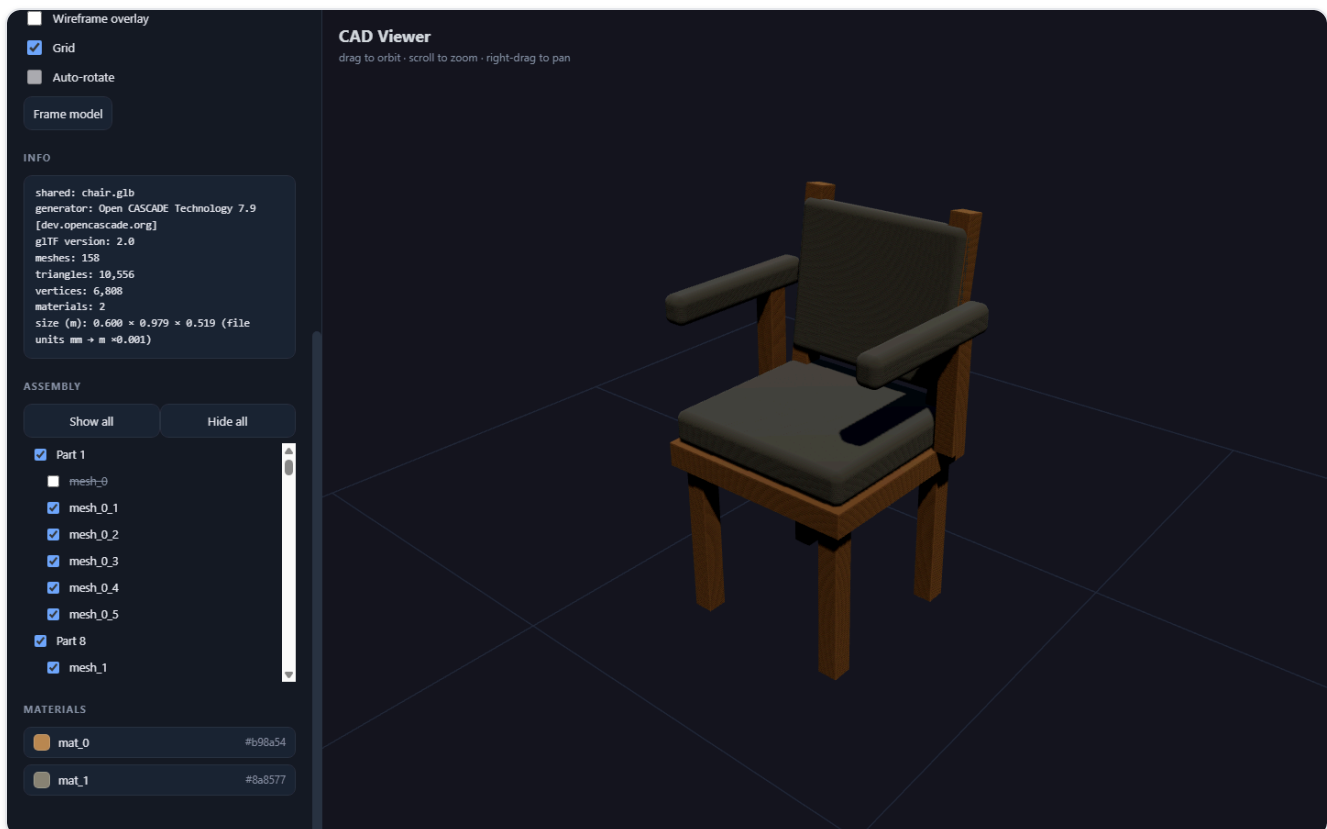
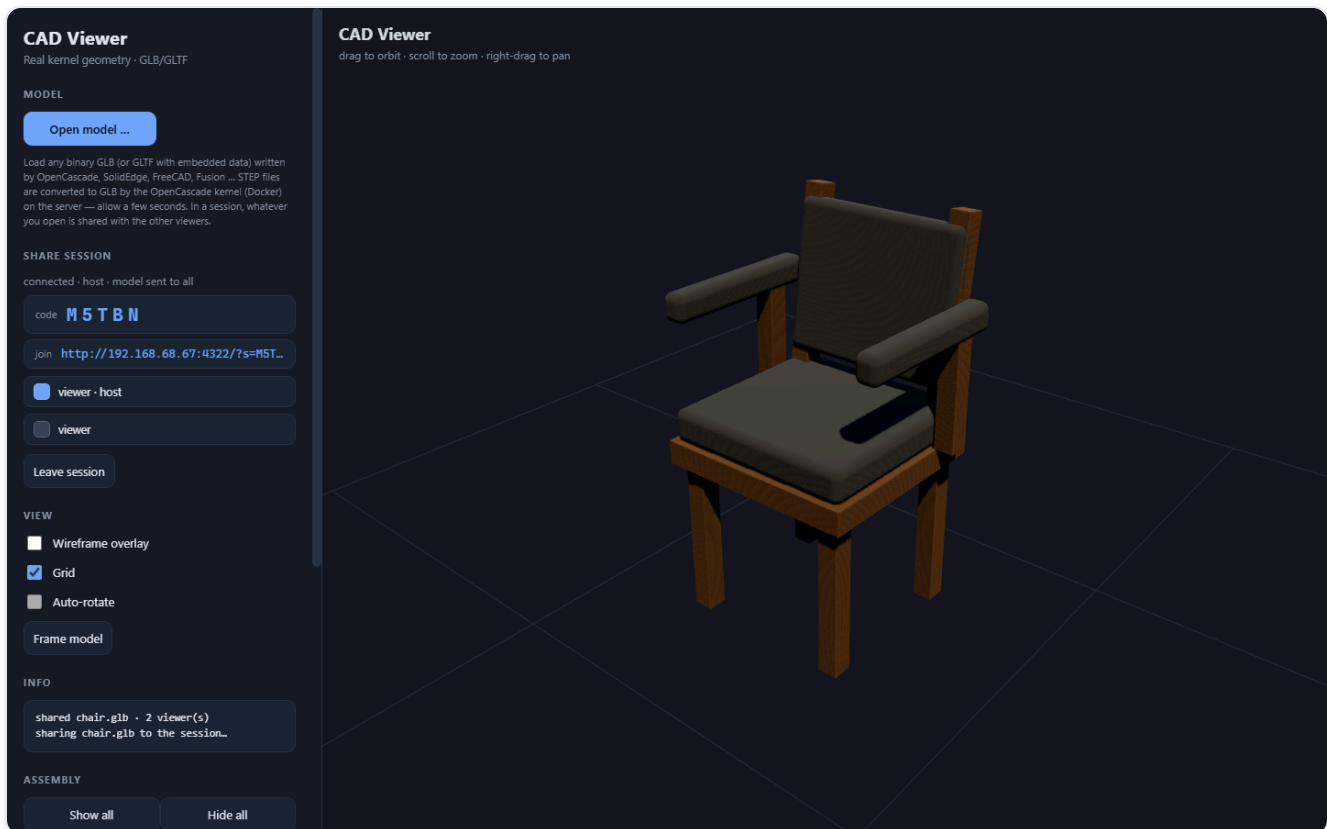
What stays in sync

- **Camera** — whoever orbits/zooms/pans, everyone follows (both directions)
- **Part visibility** — show/hide any part and it changes for all members
- **Part moves** — moving a part updates it for everyone
- **The model itself** — late joiners automatically receive the current model

and visibility state

- **Expand/collapse & selection** — the assembly-tree view and part highlight

are mirrored live across members



Security — nothing is stored on the guest. The model is **streamed to the

guest's memory only **and rendered in the browser**. No file is written to

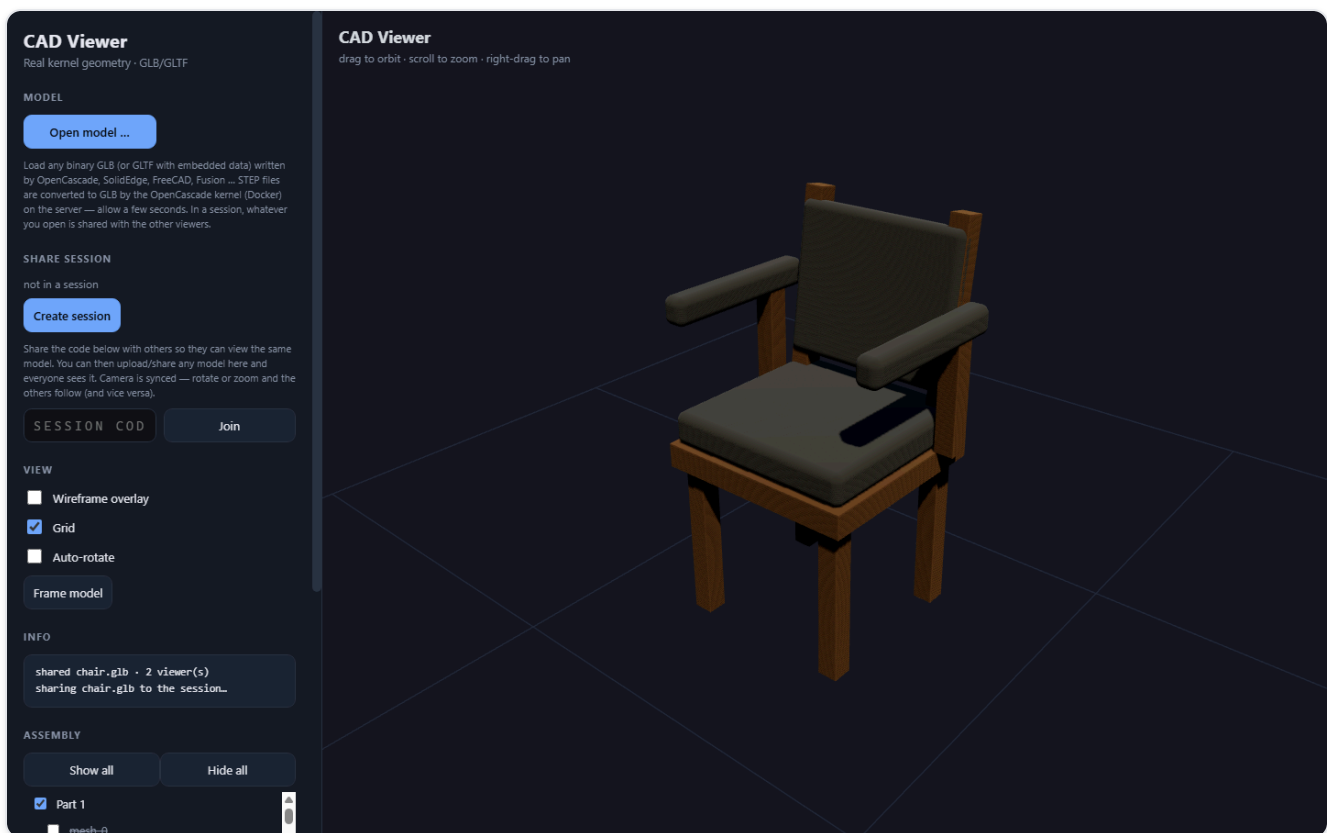
the guest's disk**; closing the tab discards it.

9. Leaving a session

Click **Leave session** to leave. What happens depends on your role:

- **Host** — you keep your local model and return to solo viewing.
- **Guest** — the shared model is **cleared from your view** when you leave on your own (for security, nothing stays on your screen), and you return to the empty viewer.

The session itself stays alive on the server for the remaining members.



10. Opening a STEP / IGES / OBJ file

STEP, IGES and OBJ conversion happens through the OpenCascade kernel (Docker).

The first time you open one you'll see the conversion overlay; when it finishes the GLB is loaded — with the original **colours and materials** preserved, and part names appearing in the Assembly tree.

OBJ colours need the .mtl file too. A Wavefront .obj stores colours in

a sibling .mtl (e.g. Asm1.obj + asm1.mtl). Select **both** files in the

Open dialog (Ctrl+click on Windows) so the materials are applied; opening

just the .obj still works, but parts get a neutral grey.

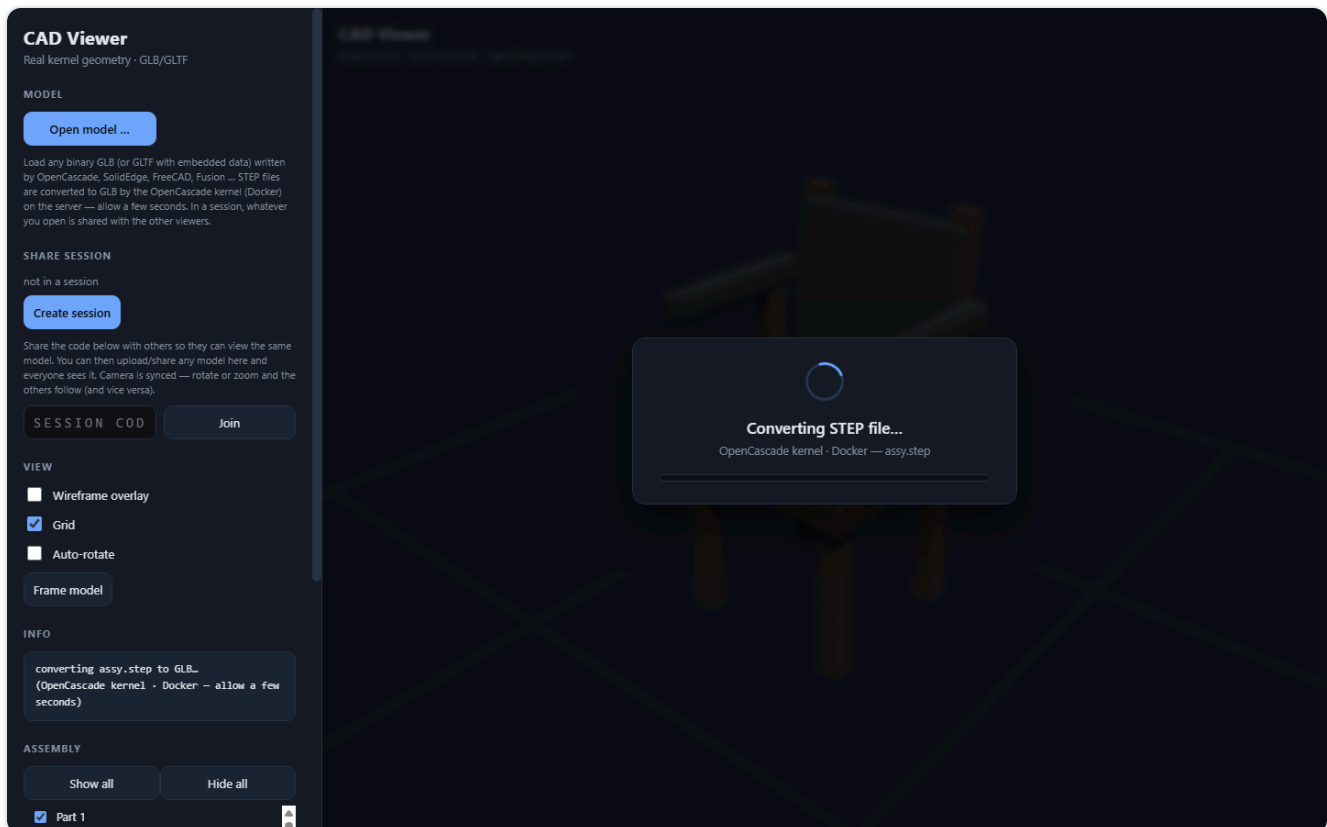
OBJ part names. Unlike STEP/IGES (which embed B-rep solid names), OBJ has

no mandatory part number. Part names come from the file's o (object) / g

(group) lines when the CAD tool writes them — e.g. SolidEdge's Asm1.obj

has none, so parts are named Part1 , Part2 . If an exporter writes o Part2

/ o Part3 , those names are used directly.



If Docker or the `chair-cq:local` image isn't installed, these formats show a conversion error while GLB/GLTF continues to work normally.

11. Standalone CAD converter (convert-cad)

Besides the viewer, the distribution includes a **separate, standalone CAD converter** — `convert-cad` (v0.1) — that turns STEP / IGES / OBJ files into GLB / GLTF **without** the viewer. It is independent of the viewer's own converters, so using it can never disturb them.

What it converts

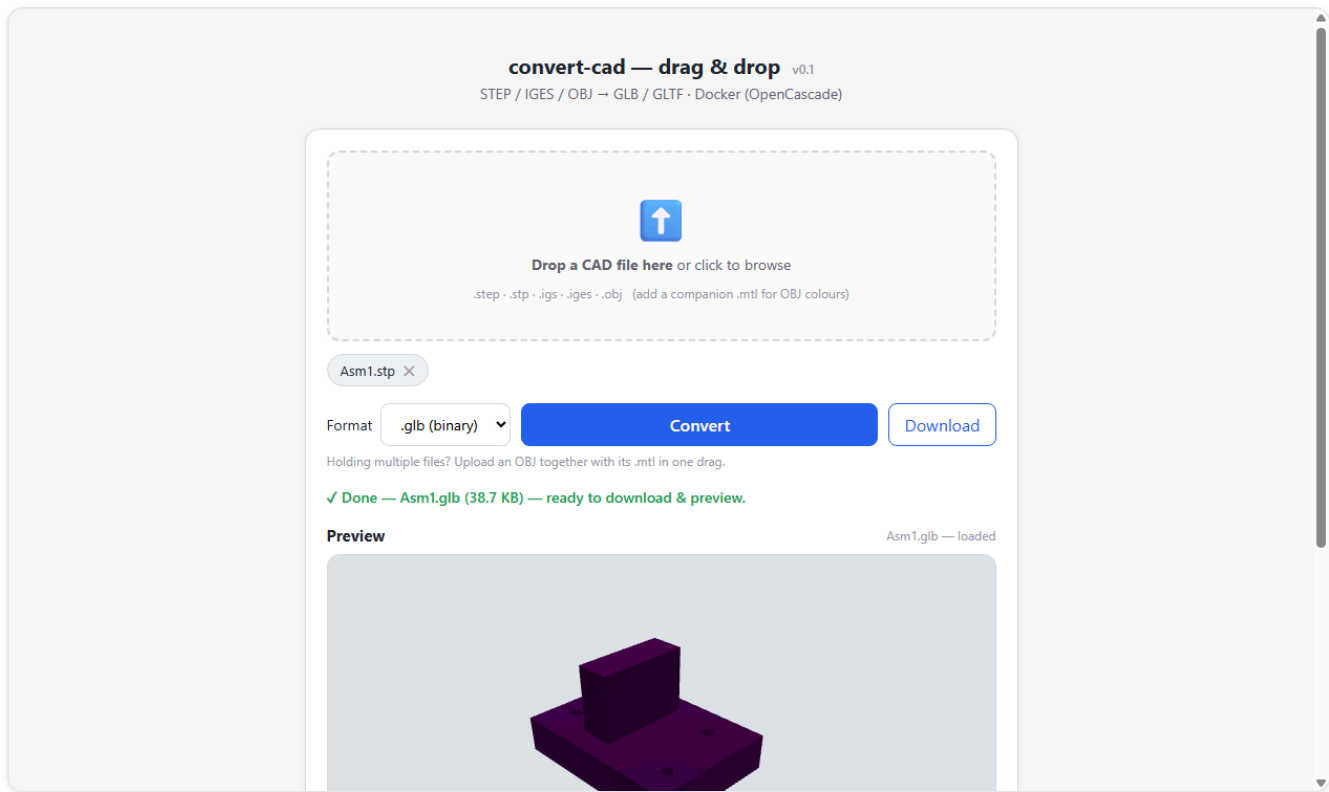
Input	Notes
<code>.step</code> , <code>.stp</code>	B-rep; keeps assembly part names + per-part colours
<code>.igs</code> , <code>.iges</code>	B-rep; keeps colours; parts named <code>Part1..N</code>
<code>.obj</code>	Mesh; part names from <code>o / g</code> lines; colours from a sibling <code>.mtl</code>

Output format is chosen by the **output file extension**: `.glb` → a single binary file; `.gltf` → text JSON + a companion `.bin`.

Using the drag-&-drop web UI (recommended)

1. Launch the web UI: double-click `convert-cad-web.bat` (Windows), or run `node convert-cad-server.mjs` from the `tools/convert-cad/` folder.
1. Open `http://localhost:8787/` in a browser.
2. **Drag & drop** a CAD file onto the page (drop an OBJ together with its `.mtl` in one drag for colours).
1. Pick `.glb` or `.gltf`, click **Convert**.
2. **Preview** the result in the built-in 3D viewer (drag = rotate, scroll =

zoom, right-drag = pan), then **Download**.



Using the command line

```
node convert-cad.mjs input.step output.glb    # or .gltf
```

Requirements

The converter runs the OpenCascade kernel inside the `chair-cq:local`

Docker image — the same one the viewer uses. Docker Desktop must be running and the image built (`install-docker-opencascade.bat` , or ``docker build -t chair-cq:local .``). **Node.js** is needed for the launcher.

How it helps sharing

Convert a CAD file to GLB/GLTF once, then hand the lightweight file to someone who opens it directly in the viewer — they never need Docker or the converter. It's the same "one compact file over the network" idea, applied offline.

12. Troubleshooting

Problem	Fix
Can't open the app on another PC	Use the join link (LAN address), make sure both PCs are on the same network, and that port 8088 isn't blocked by a firewall
STEP/IGES/OBJ shows "conversion failed"	Run <code>install-docker-opencascade.bat</code> ; confirm Docker is running with the <code>chair-cq:local</code> image
OBJ parts are grey (no colours)	The <code>.obj</code> was opened without its <code>.mtl</code> — select both files together in the Open dialog
Guest doesn't get the model	The host must be connected with a model loaded — joining an empty session shows nothing until the host shares
Port already in use	Change it: <code>start.bat 4323</code> (zip), a <code>port.txt</code> file, or <code>set PORT=4323</code>
Join link shows the wrong IP	The link uses the host's LAN address; refresh/re-create the session to re-detect it

13. Sharing with external parties

By default the viewer is meant for the **local network (LAN)**. To let someone outside your network view a session, you have two main options.

Option 1 — Port forwarding on the router

1. Find the server PC's **LAN IP** (e.g. `192.168.x.x`).
2. In your router's admin page, set up a **port forward**: forward the app's

port (default **8088**) to that LAN IP and port.

1. Share `http://<your-public-IP>:8088` with the external party.

Notes. The public IP can change (use a dynamic-DNS service if needed).

Opening a port exposes the server to the internet — use this only for trusted people, keep it temporary, and close the forward when you're done. For real deployments, prefer HTTPS (e.g. a reverse proxy like Caddy/Nginx with TLS).

Option 2 — Deploy on a public / virtual server

1. Provision a small **virtual server** (VPS/cloud instance) with a public IP

or domain.

1. Copy the distribution and run the server there (`node src/server.js`), or

package it into the same Docker image.

1. Share the server's URL with the external party.

This is the best option for **always-on** external access (client demos, vendors/subcontractors joining anytime) because it doesn't depend on a home router or a machine that must stay on.

Security reminder. Whatever route you use, anyone who has the link can view the shared model. The viewer streams the model to the guest's memory and writes nothing to disk, but for sensitive or NDA geometry, still share links only with people who are meant to see the model, and consider a private network / VPN (e.g. Tailscale, WireGuard) for the most controlled access.

CAD Viewer v0.55 — collaborative CAD viewing for the LAN.